

(2½ Hours)

[Total Marks: 75]

- N. B.: (1) **All** questions are **compulsory**.
 (2) Make **suitable assumptions** wherever necessary and **state the assumptions** made.
 (3) Answers to the **same question** must be **written together**.
 (4) Numbers to the **right** indicate **marks**.
 (5) Draw **neat labeled diagrams** wherever **necessary**.
 (6) Use of **Non-programmable** calculators is **allowed**.

1. Attempt any three of the following:

15

- Elaborate artificial intelligence with suitable example along with its applications.
- Discuss the historical evolution of Artificial Intelligence.
- State the relationship between agents and environment.
- Explain the concept of Rationality.
- Explain types of environments.
- Explain reflex agents with state.

2. Attempt any three of the following:

15

- Write the procedure for tree search.
- Explain the algorithm for breadth first search algorithm.
- Give the outline of Uniform-cost search algorithm.
- Explain A* algorithm for the shortest path.
- Give the outline of Hill climbing algorithm.
- Explain the working mechanism of genetic algorithm.

3. Attempt any three of the following:

15

- What is alpha-beta pruning? Explain the function of alpha beta pruning.
- Give the outline of min-max algorithm.
- Write a note on card games.
- What is knowledge based agent? Explain its role and importance.
- Write a note on Wumpus world problem.
- Give the outline of resolution algorithm.

4. Attempt any three of the following:

15

- What are predicates? Explain its syntax and semantics.
- What are Quantifiers? Explain the types with syntax and example.
- Convert the following into predicate form:
 - Virat is software engineer.
 - All vehicles have wheels
 - Some-one speaks some language in this class.
 - Everybody loves somebody sometime.
 - All software engineer develops software.
- Explain the process of knowledge engineering.
- What is unification? Explain the process of unification.
- Give the outline of simple forward chaining algorithm.

5. Attempt any three of the following:

15

- What is planning? Explain the need of planning.
- Explain block world problem for the following start state and end state.
- Write a note on planning graph.
- What are events? Explain its importance.
- Write a note on semantic network.
- Write a note on Truth maintenance system.